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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 07/18/2006 |
| **PI Name:** | Porter, Marc D Ph.D. | | |
| **Project Title:** | Rapid Staged Development of ISS Water Quality Sensors | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES | | |
| **Program/Discipline--Element/Subdiscipline:** | ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 50011 | **Congressional District:** | 4 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 03-OBPR-01 |
| **Start Date:** | 08/13/2004 | **End Date:** | 08/13/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Schultz, John ( Wyle Laboratories ) | | |
| **Grant/Contract No.:** | NNJ04HD71A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | This Rapid Technology Development proposal seeks to accelerate the development of Colorimetric-Solid Phase Extraction (C-SPE) technology for water quality monitoring from its current Technology Readiness Level 4 to Level 6, i.e., from a proof-of-concept, breadboard level to technology extensively tested in a microgravity environment. The product of this effort will be prototype flight hardware that will be validated by an extensive series of ground and C-9 flights. The heart of the analyzer is a diffuse reflectance spectrometer (DRS), which quantitatively measures the surface color of indicating membranes. Liquid reagents will no longer be required, only dry reagent disks, dry indicating membranes, and dry calibration tiles. The foci of the project are researching reagentless calibration, solving the air bubble entrainment in manipulating water samples in microgravity, and integrating and packaging a number of technologies into a single easy-to-use device of minimal size. We envision the endproduct to be a hand-held, reagentless, self-contained device with a built-in data system that can be applied to the onboard quantification of a | | |

variety of chemical contaminants in spacecraft water. To execute these tasks, we have assembled a team of experts from Iowa State University, the Johnson Space Center/Wyle Laboratories (JSC), and the Center for Microgravity Research/Glenn Research Center (GRC). The proposed water quality monitoring technology acceleration is distinct from yet dovetails into a recently renewed NASA project to develop C-SPE chemistries for various spacecraft water contaminants.

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

The C-SPE technique holds several distinct advantages over traditional methods in simplicity, sensitivity, and flexibility and is highly suited for application in the spacecraft environment. Several spin-off applications on Earth are envisioned, including the environmental monitoring (e.g., arsenic, lead) as well as of various substances of importance to homeland security (biowarfare agent detection).

I. Air/water separation and metering device development
Three solutions to this critical component of the project were pursued in parallel, one at each of the partnering institutions. A dual piston syringe was developed at GRC. One piston serves to remove air bubbles from water samples while the second meters exact liquid volumes through the C- SPE disk. At JSC, a liquid metering centrifuge stick has been fabricated. This device uses centrifugal force to separate air from liquid and meter the water sample. At ISU, a new C-SPE methodology capitalizing on extraction equilibrium behavior has been investigated. This approach, referred to as negligible depletion, can potentially eliminate the need for air/water separation or for accurately measuring liquid sample volumes. The following are descriptions of the progress in these three areas.

Dual Piston Syringe (GRC)

A dual piston syringe design has been developed by GRC to remove bubbles from water samples. The syringe uses a secondary piston in addition to the primary piston used in typical syringes. The secondary piston consists of a long thin shaft that passes through the center of the primary piston and has a thin stainless steel or Teflon mesh affixed to the front of the shaft to collect and push bubbles toward the exit of the syringe barrel. Pinning effects at the liquid-solid-gas interfaces and hydrophilic wetting effects are expected to preferentially let water pass through the mesh, while collecting the bubbles in the water sample. Once the bubbles are collected at the end of the syringe, both pistons are driven to expel the air. The primary piston is then used to adjust the liquid volume. The syringe is then ready for coupling to a C SPE cartridge.

Tests at 1g showed that we could remove bubbles in this manner even against gravity. While potentially attractive, this portion of the project has been discontinued owing in large part to the merits of the two other mitigation strategies, as well as budgetary constraints.

LMCS (JSC)

This portion of the project entails the development and initial ground testing of a novel approach to collect and meter a known volume of bubble-free water: a liquid metering centrifuge stick (LMCS). The LMCS is a device that utilizes centrifugal force to eliminate entrapped air and reproducibly meter an accurate volume of a liquid sample. Results obtained from ground based C SPE experiments using ionic silver, which is the biocide used by the Russian Space Agency in potable water, and either the LMCS or syringe metering, were compared to evaluate the performance of the LMCS. Although these data showed some minor differences between the two sample metering methods, they clearly illustrated the potential of utilizing centrifugal forces to achieve phase separation and meter liquid samples in microgravity.

The LMCS is based on Centrifuge-Operated Slide Staining Technology (COSST), a device previously developed for histo-chemical staining of blood smears on-orbit. Briefly stated, if a fluid sample containing bubbles in microgravity is placed in a g-field created by an on-orbit centrifuge, air/liquid phase separation will occur and entrapped bubbles will be removed from the liquid. An added benefit of this concept is that by employing a device with pre-specified dimensions, centrifugal force can also be used to collect a defined volume of liquid in preparation for analysis. Further, while the fluid is under the influence of centrifugal force, it can be directed through fluidic channels or an extraction disk.

In operation, the LMCS is charged with a water sample that may contain up to 50% of the sample volume as air. The LMCS is then loaded in a swing-out rotor of a centrifuge and exposed to a ramped g-load of 1g to 1000g. At low g-loads, air-fluid separation occurs and the water sample fills the metering cup in the center of the LMCS. Excess fluid over and above the volume contained in the metering cup is forced into the waste chamber located in the base of the LCMS via the over-flow channels. As the g load increases, the sample volume in the metering cup of the LMCS is forced through the C SPE membrane disk into the waste chamber of the LMCS. Once the centrifuge stops, the LMCS is disassembled and the C SPE extraction disk is removed and analyzed with a diffuse reflectance spectrophotometer (DRS) via the Kubelka-Munk function F(R).

It was found that LMCS provides a linear response for silver(I) in the same concentration range as the standard syringe method. There are, however, two important differences between the LMCS and the syringe data: (1) the uncertainty in the LMCS measurements is significantly higher and (2) the response at a given concentration is noticeably higher.

One likely cause for the increased uncertainty observed in the LMCS C SPE experiments is the handling of the membrane disks between centrifugation and spectrum collection. The membrane disks had to be carefully detached from the modified holder of the LMCS and placed on a standard holder prior to read-out. The transfer process could have caused minor damage to the membrane disks, leading to variability in the collected spectra.

The consistent, high response observed in the LMCS C SPE experiments is indicative of a systematic difference between the two methods that we attribute to differences in how the membrane disks are dried in the two approaches. In the standard C SPE method, a 60-mL syringe is used to force air through the membrane disks, which dries the disks before a spectrum is collected. Due to the presence of the drain holes around the metering cup in the LMCS, it is not practical to dry the membrane disks in a similar manner. As a result, the spectra of the membrane disks in the LMCS experiments were collected while the disks were still wet. Wet C SPE membrane disks are significantly more

transparent than dry disks, which is a result of reduced amount of light scattered from the membrane disk surface. The decreased scattering manifests itself as an increase in the F(R) value, which is consistent with observations from the LMCS experiments. Moreover, it is clear that light is traversing the entire wet membrane, which could result in larger light absorption due to the increased path length. Overall, this study clearly illustrates the potential of utilizing centrifugal forces to achieve phase separation and meter liquid samples in microgravity.

**Task Progress:**

Negligible Depletion (ISU)

This study has been examining the feasibility of adjusting the conditions for C SPE to eliminate the need to meter a known volume of sample through the disk. The modifications apply the principles of negligible depletion (ND) in which, beyond a predetermined sample volume, the concentration of an analyte prior to passage through the disk equals that exiting the disk. The ability to quantify analyte concentration therefore becomes independent of sample volume once achieving the condition of ND, eliminating the need to meter a known volume of sample through the disk. This capability, which would markedly simplify C SPE procedures, would be immensely attractive for deployment on Earth and in space.

Equilibration of an analyte between the liquid and solid phases in solid phase extraction follows a classic adsorption isotherm. The concentration of analyte in the stationary phase at equilibrium increases linearly with increasing analyte concentration in the original sample, gradually reaching a plateau. A comparable situation is encountered in C SPE. That is, with a fixed volume of sample, a plot of F(R) at first increases linearly with increasing concentration, and eventually levels out.

A plot with a similar shape is obtained in C SPE when a water sample of a fixed concentration, but with varied sample volume, flows through an extraction disk. In this case, a plot of F(R) against volume also undergoes an initial linear increase. Further increases in volume eventually result in the equilibration of the analyte between the two phases such that the concentration of analyte leaving the disk is the same as that entering the disk. This situation is signaled by a plateau in the F(R) response, which is diagnostic of the condition of ND. The height of this plateau is a function of concentration.

In earlier work, iodine in water was selectively determined with C SPE at low levels (limit of detection of 0.05 ppm) by accurately passing a 10.0 mL sample through an Empore disk. In testing the performance of this system, the results showed that achievement of the condition of ND required more than 20 mL of sample, twice that used in conventional C SPE.

To minimize the amount of sample required, several other materials were surveyed as possible extraction media. Of the tested materials, nylon filters impregnated with PVP extracted iodine efficiently and at much smaller volumes. We found, for instance, that nylon disks impregnated with a 1.0% PVP solution yielded a much larger value of F(R) after iodine extraction than that obtained for Empore disks impregnated with a 3.0% PVP solution. We attribute this difference to a higher level of PVP-loading with the nylon disks.

As noted, the survey also showed that the volume of sample required to achieve the condition of ND with the PVP-loaded nylon media was less than that needed for the impregnated Empore material. Microscopic examination revealed that the nylon disks have a much smoother surface and greater material density than the Empore disks. This difference suggests that the lower sample volume necessary for operation in an ND mode with nylon arises from a smaller sampling depth for the diffusely reflected light. It therefore appears that nylon disks have the potential to perform effectively as an extraction medium for the determination of iodine by C SPE/ND.

The results of experiments designed to determine the sample volume required for ND using nylon membranes impregnated with PVP showed that the value of F(R) becomes effectively constant between 6.0 and 9.0 mL. Based on these data, the determination of iodine by C SPE/ND can be easily carried out in ~60-70 s by: 1) quickly filling a syringe with any volume of sample, as long as it surpasses 9.0 mL; 2) passing this sample through the extraction disk; and 3) reading the disk using the DRS.

In conclusion, the thin nylon disks used here provide fast extraction kinetics and a high concentration factor for the analytes investigated. The small sampling depth of the disks enables ND to be attained after a relatively small sample volume. The implications of the ability to perform SPE or C SPE under conditions of ND are enormous. The capability to determine a target analyte in solution without measuring the amount of sample means that an entire step has been eliminated from the general analytical process. This innovation now opens the way for even more facile applications of C SPE when operated in conjunction with ND. That is, C SPE/ND is now a two-step process - extraction and disk readout - and not the collection of an accurate sample volume, extraction, and disk readout. Efforts to extend this concept to a range of analytes with both ground and space-bound relevance are underway. Approaches for the determination of silver(I), which uses an impregnated colorimetric reagent, will likely resemble that employed herein for iodine. We are also examining ways in which methods for analytes like nickel(II), which rely on a precipitation reaction, can be adapted to function in an ND mode.

II. Data system and software

The plan set forth in the proposal called for the phased development of the data system and software from the Phase 1 configuration consisting of a BYK-Gardner DRS for data acquisition, interfaced to a laptop computer for data analysis, through a Phase 2 system incorporating a PDA to replace the laptop, to a fully integrated single instrument in Phase 3 containing all functionalities. Discussions with BYK-Gardner have led to the formulation of an alternative plan that bypasses Phase 2, which will not only be less costly and be implemented more quickly, but will also provide a data system with most of the important functions. The only function omitted in this plan is the barcode reader. In this approach, all of the data analysis functions performed with the laptop will be implemented in the DRS through changes in the instrument firmware. These changes will be developed by BYK-Gardner and provided in the form of a firmware upgrade that can be applied to existing hardware. Wyle team members have drawn up a preliminary statement of work to be executed by BYK-Gardner for the desired data system and software.

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